Research on WiFi CSI Gesture Recognition Technology and Application Prospects

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Abstract. With the rapid development of 5G and IoT technologies, contactless human-machine interaction technology has ushered in significant opportunities. WiFi Channel State Information (CSI) serves as an emerging wireless sensing method, enabling precise gesture recognition without the need for additional hardware support. It offers advantages such as low deployment costs, strong privacy protection, and high environmental adaptability. This paper reviews the current mainstream WiFi CSI gesture recognition technologies, focusing on the technical principles and performance of representative methods such as AirFi, Wi-AM, Multimodal Fusion GMM, WiHF, and Cross-domain Gesture. Additionally, it delves into the key challenges faced in the field, including environmental dependency, high data costs, insufficient real-time performance, and user variability, and proposes corresponding solutions. Finally, it explores the future development potential and research trends of WiFi CSI gesture recognition in practical application scenarios such as smart homes, healthcare, and gaming and entertainment. This paper systematically reviews the research progress and challenges in WiFi CSI-based gesture recognition technology, providing a technical development framework and future research directions for the field.

Keywords: WiFi sensing, Channel State information (CSI), Gesture Recognition, Contactless Interaction, Multimodal fusion.

1 Introduction

In daily life, gestures are a natural and efficient means of communication. With the continuous development of human-computer interaction technology, how to enable machines to recognize and understand human gestures has become an important topic in the design of intelligent systems. Traditional gesture recognition methods primarily rely on cameras or wearable devices. While these methods offer high accuracy, they also pose privacy risks, are highly dependent on specific devices, and are sensitive to environmental lighting conditions, making them difficult to meet the flexibility and universality requirements of practical applications.

In recent years, gesture recognition technology based on WiFi signals has gained increasing attention. This technology analyzes the minute changes in signal propagation

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caused by human movements, extracts gesture features, and performs recognition. Compared to visual or wearable solutions, WiFi sensing does not require users to wear devices and is not restricted by obstructions or changes in lighting, making it easier to integrate into various real-world scenarios such as homes, offices, and medical settings.

Among these, Channel State Information (CSI) is currently one of the most widely used wireless sensing data types. It can be obtained from commercial WiFi chips and features high granularity and sensitivity, making it highly suitable for gesture sensing. CSI technology not only reduces the deployment costs of sensing systems but also offers superior solutions for privacy protection, thereby attracting significant attention in both academic and industrial circles.

This study systematically reviews current mainstream WiFi CSI gesture recognition methods, analyzes their technical frameworks and core concepts, and discusses their performance based on experimental data. Additionally, this paper summarizes the main challenges in this field and explores future development directions in areas such as multimodal fusion, model optimization, and application expansion.

2 Typical technology comparison and analysis

As research into the application of WiFi CSI in gesture recognition continues to deepen, academia and industry have proposed a variety of technical solutions to key issues in non-contact interaction applications, such as environmental robustness, data efficiency, real-time response capabilities, and user variability adaptation. These solutions not only reflect the technological evolution of core perception mechanisms but also highlight the process of WiFi CSI gesture recognition moving from theoretical exploration to practical deployment.

2.1 AirFi：enabling environmental generalization with adversarial encoders

The introduction of the AirFi system marks a key breakthrough in the application of adversarial learning methods to CSI gesture recognition. By incorporating an adversarial autoencoder (AAE) structure and utilizing maximum mean squared distance (MMD) for distribution alignment, the system effectively filters out environment-related noise interference and extracts stable, environment-independent features. Similar approaches can be traced back to SimGAN, which employs adversarial training to enhance the realism of synthetic images, inspiring the application of “domain adversarial” techniques in the wireless signal representation space. Another prior study, Domain-Adversarial Neural Network (DANN), first systematically introduced gradient reversal layers (GRL) for domain-agnostic learning in image classification tasks. This idea was effectively adapted by AirFi for adversarial modeling of CSI features, enhancing the system's generalization capabilities across multiple scenarios [1].

2.2 Wi-AM：residual networks combined with meta-learning to achieve cross-domain adaptation

Wi-AM not only combines adversarial training with residual neural network structures, but also introduces a Model-Agnostic Meta-Learning (MAML) strategy to enable the model to quickly adapt to new environments or users. This method inherits the domain-agnostic feature extraction concept from Domain-Adversarial Training (DAT) and combines it with ResNet's hierarchical representation advantages in processing multi-scale Doppler spectra, significantly enhancing the model's feature expression capabilities. Further research, such as MetaSense, adopts an analogy-based learning approach, leveraging rapid in-task adjustment mechanisms to enhance adaptation speed in indoor spaces with severe channel variations. Additionally, Wi-AM benefits from the light-weight gradient update mechanism proposed by the Reptile algorithm, making it suitable for resource-constrained edge devices. The system demonstrates robustness far exceeding traditional transfer learning methods in cross-environment deployment, particularly suited for smart IoT systems where gesture recognition models frequently need to be transferred to new devices or regions [2].

2.3 Multimodal Fusion GMM：multiple receivers + statistical modeling enhance robustness

Multimodal Fusion GMM adopts a spatial multi-receiver design concept, collecting CSI information from multiple paths and performing singular value decomposition (SVD) to extract dominant components, thereby enhancing the stability of channel spatial features. Additionally, filter fusion (MFB) further removes background interference to extract more robust dynamic gesture features. A system with a similar technical approach is Widar3.0, which simulates a multi-receiver structure through beamforming to improve spatial resolution. Furthermore, FusionNet introduces a Bayesian-weighted multimodal fusion mechanism, enabling high-confidence recognition while ensuring information redundancy. Multimodal Fusion GMM employs a Gaussian mixture model (GMM) in the classification stage to avoid overfitting and high latency issues associated with deep learning models, making it suitable for edge computing or low-power applications. This method performs exceptionally well in medical monitoring and smart classrooms with strong multipath reflections and complex environments [3].

2.4 WiHF:dual-task design that balances personalization and real-time functionality

WiHF achieves parallel optimization of gesture recognition and user identification tasks by introducing a dual-task structure. The main network uses a lightweight CNN to extract spatial features, while GRU captures temporal dynamics. The auxiliary task supervises user identity based on shared feature layers, guiding the model to focus on individual differences. This strategy balances personalization and generalization. A similar approach is also reflected in the DeepSense framework, which achieves multi-

task collaborative optimization of user perception and behavior perception by integrating multi-sensor modalities. Additionally, the DFS spectrum input features of WiHF undergo Seam Carving preprocessing, significantly compressing redundant dimensions and reducing model latency to sub-10ms levels, providing real-time guarantees for time-sensitive scenarios. Subsequent systems like IDNet-Gesture further enhance the discriminative power of user embedding vectors, providing theoretical support for the deep integration of user branches in WiHF. The WiHF system has become the technological foundation for personalized intelligent interaction in scenarios such as home automation, industrial control, and smart access control [4].

2.5 Cross-domain Gesture：densenet + dynamic convolution enhances generalization ability

The Cross-domain Gesture system aims to address the issue of significant performance fluctuations in WiFi gesture recognition across multiple directions, devices, and environments. This method utilizes the DenseNet architecture, which retains low-level details and high-level semantic information through its dense connection mechanism, thereby enhancing the model's ability to extract fine-grained action features in complex contexts. To further improve generalization, the study introduces a Dynamic Convolution module, which can adaptively adjust convolution kernel parameters based on input features, enabling the model to maintain stable responses even when faced with different scene changes (such as user posture, angle, or furniture obstruction). This dynamic adjustment mechanism is similar to the conditional channel attention concept in MobileNetV3, enhancing the model's ability to perceive environmental changes. Additionally, Cross-domain Gesture integrates a multi-task loss function to assist in learning direction classification tasks, thereby enhancing the main task's ability to discern spatial structural changes. Experimental results demonstrate that the system maintains an identification rate of over 98% across multiple unknown room deployments, making it suitable for applications requiring high robustness, such as virtual reality interaction, multi-terminal collaboration, and intelligent meeting control [5].

2.6 RoGER：small-sample multimodal recognition framework

RoGER achieves high-precision gesture recognition under extremely low-sample conditions by integrating modal transfer mechanisms with few-shot learning strategies. In addition to borrowing Prototypical Networks to construct class centers, RoGER also combines Relation Networks to enhance inter-class discriminative power by learning similarity functions between samples. In terms of modal transfer, it uses a shared encoder structure to uniformly map different modalities and fuses complementary information during the decoding stage, effectively overcoming the issue of modal distribution differences. Similar ideas are also found in the Cross-modal Meta-Learning (CMML) model, which enhances cross-modal transfer performance by jointly embedding spaces in image-text matching. Additionally, the attention-matching module in RoGER is inspired by the FSL-DA framework, enabling dynamic emphasis on key modal segments to further improve recognition accuracy and noise resistance. This

technology provides feasible pathways for emergency deployment, human-machine interfaces, and unsupervised modeling [6].

2.7 CUDSG: GAN enhances generalization and anti-interference capabilities

CUDSG leverages Generative Adversarial Networks (GANs) to introduce a pseudo-data augmentation mechanism into traditional CSI recognition systems, providing robust recognition capabilities in real-world scenarios where user samples are insufficient and environmental changes are frequent. Another approach aligned with CUDSG is Gesture-Aware GAN (GAGAN), which enhances action category discrimination through conditionally guided generation using specific labels. AugGAN, on the other hand, focuses on addressing sample imbalance issues by generating high-quality CSI sequences for minority classes. CUDSG further enhances this by introducing multi-scale discriminators to model features across different time windows, making the generated samples more credible in terms of temporal characteristics. Additionally, CUDSG combines GAN-based domain generalization strategies to generate approximate scene features without target domain data, breaking the traditional cross-domain method's reliance on target data. This model provides robust support for cross-user and cross-device recognition tasks in open environments and lays the foundation for future research on the safety of adversarial training [7].

2.8 HGR System: lightweight implementation of traditional machine learning methods

The HGR system returns to the traditional machine learning paradigm, emphasizing the ability to achieve efficient, deployable gesture recognition even under conditions of limited computing resources. In addition to PCA and FDA feature extraction strategies, HGR also references LDA (linear discriminant analysis) and ICA (independent component analysis) combined dimension reduction techniques to enhance signal independence and improve the model's ability to distinguish between multiple users. Similar systems, such as Wisture use decision trees and clustering algorithms for lightweight recognition, and can even run on a single-core MCU, demonstrating the practicality of traditional models on embedded platforms. Additionally, the SVM classifier in the HGR system uses a radial basis function (RBF) to handle nonlinear features, retaining nonlinear decision-making capabilities without increasing network complexity. This solution has broad application value in scenarios such as education and security that have explicit requirements for low latency and high interpretability [8-10].

3 Key challenges and response strategies

Although WiFi Channel State Information (CSI) gesture recognition technology has made significant progress in recent years, its advantages, such as low cost, non-intrusive operation, and wall-penetrating recognition, have demonstrated broad application prospects in fields such as smart interaction and human-machine interfaces. However,

during actual deployment, it still faces multiple challenges that hinder its large-scale promotion and stable operation.

Firstly, strong environmental dependency is the core issue limiting the generalization ability of CSI models. Since wireless channels are highly sensitive to physical space, factors such as wall layout, reflection interference, and human movement can cause significant changes in channel characteristics, leading to a significant decline in model accuracy when migrating between different environments. Research has shown that the accuracy difference between the same model in open-plan home scenarios and closed-off office spaces can exceed 18%. To address this issue, the AirFi system proposes using adversarial autoencoders (AAE) to extract environment-independent representations and introduces maximum mean squared distance (MMD) to align the distributions of the source domain and target domain, significantly enhancing the model's cross-scenario robustness. Building on this, Wi-AM further combines domain discriminators with meta-learning mechanisms, using adversarial training to enable the model to learn domain-agnostic shared features and quickly adapt to new scenarios, effectively mitigating performance fluctuations caused by domain shifts. Additionally, CUDSG uses generative adversarial networks (GANs) to synthesize diverse synthetic samples, expanding the training dataset distribution across user and scenario dimensions, providing a generative solution to environmental adaptability issues.

Second, the high cost of data collection limits the flexible deployment of CSI gesture recognition in real-world applications. Especially when new environments or new users are introduced, deep learning models often require the collection of a large amount of annotated data (e.g., the Multimodal Fusion GMM system requires thousands of gesture samples for modeling), significantly increasing the deployment threshold of the system. To address this, Wi-AM introduces a meta-learning algorithm based on MAML, enabling the model to achieve rapid fine-tuning with minimal samples while maintaining strong initialization capabilities, significantly reducing training overhead. The RoGER system, meanwhile, employs a few-shot learning strategy and modal shared space construction, requiring only dozens of data points to complete feature transfer and recognition tasks across different modalities, making it suitable for early product testing, temporary deployment, and other scenarios with limited data resources.

Third, real-time performance is another critical challenge that deployment-based systems must address. Traditional CNN and RNN architectures have high computational complexity when processing CSI spectrum plots and time-series features, making it difficult to meet the millisecond-level response speed required by real-time interactive applications such as AR/VR and smart home control. To address this, the WiHF system employs the Seam Carving algorithm for feature compression in its model structure, combined with a lightweight CNN and GRU architecture, significantly reducing model runtime and keeping overall processing latency below 10 milliseconds, thereby meeting the requirements of high-response scenarios. Similar approaches, such as the IDNet-Gesture system, also incorporate feature pruning and weight pruning strategies within a multi-task framework, enabling the model to achieve low-latency operation on edge devices.

Finally, individual user differences are one of the key factors affecting system recognition stability. Due to natural variations in speed, amplitude, and direction when individuals perform the same gesture, models lacking personalized capabilities may experience reduced recognition accuracy or even misjudgments. To address this issue, WiHF employs a dual-task neural network architecture, where the primary task performs gesture classification and the auxiliary task conducts user identity recognition. By sharing the underlying feature structure, the model simultaneously learns action features and captures user behavioral deviations, effectively enhancing individual adaptability. Experimental results show that its user recognition accuracy reaches 96.74%, demonstrating high practicality in applications such as distinguishing family members and identity-aware control.

In summary, addressing the four major challenges of environmental adaptability, sample efficiency, response speed, and individual differences, current research has proposed various strategies, including adversarial learning, meta-learning, few-shot modeling, multi-task collaboration, generative augmentation, and model lightweighting, laying a solid theoretical foundation and system architecture support for WiFi CSI gesture recognition technology to transition from the laboratory to real-world applications. Future work could further explore model interpretability, security, and cross-device collaboration capabilities to achieve truly scalable and maintainable ubiquitous sensing systems.

4 Conclusion

WiFi CSI gesture recognition technology, as a new generation of contactless sensing method, has demonstrated broad application potential and technical advantages in multiple fields such as smart home, medical monitoring, and virtual reality in recent years. Compared with traditional visual or wearable solutions, this technology has made significant progress in terms of recognition accuracy, environmental adaptability, and user-friendliness. Current mainstream methods such as AirFi, Wi-AM, and WiHF have achieved good results in multi-domain generalization, real-time performance, and user difference modeling, driving gesture recognition from experimentation to practical application. However, issues such as strong environmental dependency, high demand for training samples, and individual user differences remain significant bottlenecks limiting its large-scale deployment and commercialization.

Looking ahead, research and development in WiFi CSI gesture recognition can be explored in multiple directions. First, multimodal fusion will become a trend, combining perception technologies such as UWB and millimeter-wave to further enhance the system's spatial feature resolution and interference resistance. Second, edge computing optimization will accelerate the recognition process through model compression (such as knowledge distillation) and hardware collaboration, meeting the demand for low-latency real-time interaction. In terms of privacy protection, distributed training strategies such as federated learning will facilitate localized processing of user data, reduce data transmission risks, and enhance system compliance. Additionally, establishing a

unified standardized ecosystem to standardize data collection, model training, and evaluation processes will help lower system development barriers and accelerate technology adoption. Finally, enhancing the model's adaptability to complex interaction scenarios involving multiple users, tasks, and locations will be key to driving the widespread application of this technology.

Authors Contribution. All the authors contributed equally and their names were listed in alphabetical order.

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